

SeaRAG: Reducing Hallucination in Retrieval-Augmented Generation via Statement-Entity Adaptive Ranking

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Abstract

Retrieval-augmented generation (RAG) has facilitated large language models (LLMs) by grounding facts in external knowledge. While prior RAG strategies can achieve dynamic control by detecting sentence-wise hallucination, thereby reducing unnecessary computation, they ignore the entity, statement, and their combinations. In this work, we analyze adaptive RAG methods through probing uncertainty to explore real-time entity- and statement-level verification, finding that such information can serve as training-free signals for hallucination detection, and the entropy probes can also guide principled document ranking. Inspired by these insights, we propose SeaRAG, a training-free, adaptive RAG framework that dynamically detects and mitigates hallucinations by probing LLMs at both entity and statement aspects, ranking retrieved passages via entropy-based uncertainty reduction, and regenerating evidence-grounded responses in real-time. Experiments across various Question-Answering benchmarks and multiple LLMs demonstrate consistent accuracy improvements and reduced retrieval rates. SeaRAG with self-correction outperforms Always Retrieve by up to 11.1% on TriviaQA while cutting retrieval frequency from 45.2%, offering an efficient and real-time hallucination control.

CCS Concepts

• Computing methodologies → Natural language generation.

Keywords

Retrieval-Augmented Generation, Large Language Models, Hallucination Detection, Dynamic Retrieval

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1 Introduction

Retrieval-Augmented Generation (RAG) has emerged as a powerful paradigm for enhancing Large Language Models (LLMs) by grounding their responses in external, verifiable knowledge sources [2, 12, 19, 23, 35, 38]. Unlike static parametric knowledge stored during pretraining, RAG enables models to incorporate up-to-date and domain-specific information dynamically, thereby improving factual accuracy and reducing hallucinations [5, 30, 31]. This makes RAG particularly valuable for applications such as question answering, fact verification, and knowledge-intensive reasoning, where reliability and timeliness are critical.

Despite these advantages, current RAG approaches stumble around notable challenges. Most RAG systems adopt the "retrieve-for-all", which always retrieves documents for all queries, regardless of whether the LLM already has sufficient knowledge within itself. [15, 33] This practice introduces unnecessary computational overhead, increases inference latency, and may even harm accuracy when irrelevant or noisy passages are injected. Efforts to introduce adaptive retrieval commonly rely on supervised training or specialized estimation modules [4, 33], which demand significant data and engineering resources while offering limited interpretability. Meanwhile, hallucination control is often post-hoc, detecting errors after generation rather than preventing them, and many detectors require retraining that does not generalize across tasks or LLMs.

In this work, we investigate adaptive RAG through the lens of active uncertainty elicitation and evidence selection. We investigate two research questions: (1) *Can real-time probing of the LLM—posing verification questions about entities and statements—serve as an effective, training-free signal for detecting hallucinations during generation?* (2) *Given a pool of retrieved candidates, can entity-level and statement-level entropy measured from the LLM's responses provide a principled basis for ranking passages, thereby improving grounding efficiency and factual reliability?* Empirical results demonstrate both hypotheses. Real-time probing yields robust, interpretable signals that identify when the model is uncertain about specific entities or the overall claim, enabling selective retrieval rather than always

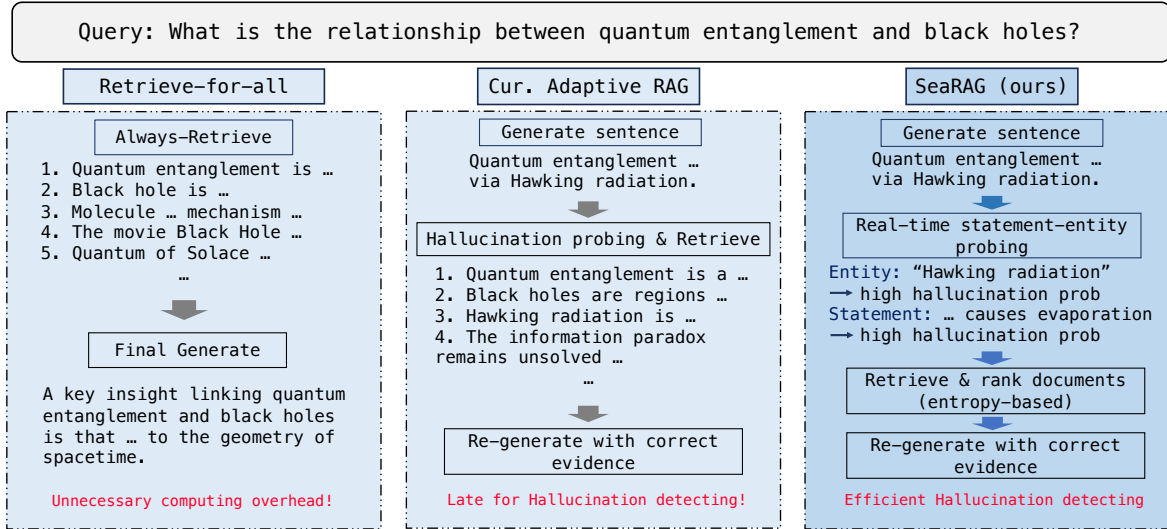


Figure 1: Comparison among different RAG frameworks for the same query. "Retrieve-for-all" retrieves some unnecessary documents, while current adaptive strategies generate hallucinations only from the sentence perspective, and detect them too late. The proposed SeaRAG detects high confidence both entity-wise and statement-wise, skipping redundant retrieval.

retrieving. Furthermore, entropy-based ranking consistently prioritizes evidence that directly resolves the probed uncertainties, reducing reliance on noisy or tangential context and improving factual consistency without additional training overhead.

Inspired by the above, we propose a Statement-Entity Adaptive Ranking for RAG, named SeaRAG, which is a unified adaptive framework that integrates real-time probing with entropy-based ranking. A probing detector first interrogates the model at two complementary levels: entity-level, by extracting named entities from the generated sentence and asking targeted verification questions; and statement-level, by eliciting an explicit confidence judgment about the overall claim. From the model’s responses, we compute multi-signal hallucination scores that combine response entropy, token-level confidence, and linguistic markers of uncertainty. When retrieval is warranted, the system gathers candidate passages. It ranks them using entropy measured from passage-conditioned probes at both levels: entity explanations in context and Yes/No consistency checks for the claim. The top-ranked passages—those that most decisively reduce uncertainty—are incorporated into the prompt, and the sentence is re-generated with evidence, yielding an end-to-end, training-free pipeline that both detects and mitigates hallucinations in real-time while minimizing unnecessary retrieval.

We validate SEA RAG on knowledge-intensive QA benchmarks and diverse model backbones. Experiments show that probing-guided retrieval reduces unnecessary calls without sacrificing accuracy, and entropy-based ranking improves grounding quality by selecting highly relevant passages that align with the probed uncertainties. Our method achieves 66.71% accuracy on TriviaQA, outperforming Always Retrieve (55.59%) by 11.12% while reducing the retrieval rate from 100% to 45.2%. The approach demonstrates consistent improvements across all datasets: PopQA (35.96% vs 31.04%), Natural Questions [22] (33.45% vs 25.13%), HotpotQA [40] (36.30%

vs 27.20%), and 2WikiMQA (39.15% vs 32.21%). Ablation studies confirm the complementary roles of entity-level and statement-level probing, while analyses of retrieval frequency and correction behavior demonstrate the efficiency and stability of the proposed control loop. In summary, this paper makes 4 contributions:

- We introduce a training-free detector that actively probes the LLM at entity and statement levels, producing interpretable hallucination signals from response entropy, confidence, and uncertainty markers during generation.
- We propose an entity- and statement-level entropy criterion to rank retrieved candidates by how decisively they resolve probed uncertainties, selecting evidence that maximally improves factual grounding.
- We integrate probing-based detection and entropy-based ranking into a single inference pipeline that retrieves only when needed and re-generates with top-ranked evidence, reducing hallucinations with minimal overhead.
- We demonstrate consistent gains in factuality and retrieval efficiency across benchmarks and models, with analyses showing robust behavior under varying retrieval budgets and generation settings.

2 Related Work

The Retrieval-Augmented Generation (RAG) paradigm has become essential in enhancing large language models (LLMs) by integrating external knowledge, improving response accuracy, and reducing hallucinations [3, 8]. A critical advancement in this domain is the development of adaptive RAG systems that dynamically control retrieval depth, scope, and iteration based on the complexity and evolving needs of user queries.

2.1 Adaptive Retrieval Strategies in RAG

Recent work has explored how to make retrieval-augmented generation more efficient by adapting whether and how much to retrieve external knowledge, instead of always using retrieval for each query. Adaptive-RAG [15] employed a smaller classifier to predict the complexity of an input question to choose between no retrieval, single-step retrieval, or more complex multi-step retrieval strategies. PAIRS [4] forced the model to generate a direct answer and a context-augmented answer, when these two answers were consistent, retrieval could be skipped; otherwise, retrieval would be activated, and then documents were filtered by an adaptive selection module. Dynamic prompting RAG [10], SAM-RAG [43], and KBAAlign [42] explored retrieval depth and scope based on evolving context. SUGAR [44] determined retrieval time and steps by calculating context semantic entropy, SEReDeEP [36] employed semantic-entropy probes for hallucination detection, and BEE-RAG [37] achieved on long-context RAG through attention entropy. These methods substantially reduce retrieval costs for simpler queries while retaining or improving accuracy. However, these works did not directly integrate mechanisms for controlling hallucinations during generation.

2.2 Hallucination Detection and Control

Hallucination—fluent but factually incorrect or unsupported content—remains a major concern in LLMs [29]. [5] fine-tuned "small" LMs to output "I don't know" to reduce hallucination during the RAG process. Semantic Entropy Probes (SEPs) [21] approximated semantic uncertainty from hidden states of a single generation, rather than relying on multiple sampled generations, and show strong generalization to out-of-distribution tasks. HARP [13] decomposed hidden state spaces into semantic vs reasoning subspaces, and uses projections onto the reasoning subspace as features for detection. IoE prompting [24] identified the "confidence" of LLMs during the self-correction process, which they consider may result in unreliable conclusions. RAG-HAT [31] trained hallucination detection models that generate detection labels and provide detailed descriptions. Obeso et al. [27] presented a real-time identification of hallucinated tokens in long-form generations. Nonetheless, these approaches might struggle in combination with adaptive retrieval during the RAG process.

2.3 Uncertainty-Based Signals

Employing entropy or internal uncertainty as a signal for model decisions has received growing attention. Dhole [7] designed a strategy that performs retrieval augmented generation with dynamic retrieval through uncertainty detection. Yarie et al. [41] aimed to reduce token-level uncertainty by modifying the LLM architecture to incorporate a token-level uncertainty estimation layer. UncertaintyRAG [25] exploited Signal-to-Noise Ratio (SNR)-based span uncertainty to estimate similarity between text chunks. Soudani et al. [32] presented an axiomatic framework to identify deficiencies in existing methods and guide the development.

Despite these advances, prior RAG methods either retrieved indiscriminately or relied on specialized modules, and most hallucination control techniques are post-hoc and model-specific, leading to inefficiency and limited generalizability for broader domains.

There is no training-free, unified approach that can both detect hallucinations in real-time and select relevant evidence tailored to the model's specific uncertainties.

3 Method

We propose an adaptive RAG system, which combines real-time probing of LLMs with entropy-based ranking of retrieved evidence. Unlike previous probing methods that infer uncertainty from token probabilities, our approach actively interrogates the model to detect hallucinations at both entity-wise and statement-wise, as shown in Figure 2. When potential hallucinations are recognized, we retrieve candidate documents and rank them according to entropy to select the most relevant evidence with less hallucination. This dual-stage design—active detection followed by intelligent ranking, aiming to minimize hallucinations while maintaining efficiency.

3.1 Problem Formulation

Given a user query $x = (x_1, \dots, x_T)$, a causal language model f_θ with parameters θ produces a sequence of hidden states $h_t \in \mathbb{R}^d$ at each position t and generates output tokens through a distribution $p_\theta(y_t | x, y_{<t}) = \text{softmax}(W_o h_t)$, where $W_o \in \mathbb{R}^{|\mathcal{V}| \times d}$ is the output projection matrix and $\mathcal{V}$ is the vocabulary. A retrieval system optionally provides a set of passages $\mathcal{C} = \{c_1, \dots, c_K\}$ from an external corpus, which can be incorporated into the generation context to ground the model's outputs in factual evidence.

Our objective is to minimize hallucinations—factually incorrect or unverifiable statements—while preserving generation fluency and efficiency. We formalize this as a joint optimization problem where the system must decide when to retrieve ($\delta_t \in \{0, 1\}$ at each generation step t), which passages to use from retrieved candidates ($\mathcal{C}_{\text{selected}} \subseteq \mathcal{C}_{\text{retrieved}}$), and how to incorporate them into generation. Traditional approaches either retrieve for all queries (high cost, potential noise) or use heuristics that may miss genuine uncertainty. Our approach learns to detect hallucinations through active probing—querying the model itself about its confidence—and selects evidence through entropy-based ranking, which measures how well each passage addresses the specific detected uncertainties. This formulation enables fine-grained, adaptive retrieval that triggers only when the model exhibits genuine factual uncertainty, as revealed through its responses to verification questions, and ensures that only the most relevant evidence is incorporated.

3.2 Overall Architecture

Our pipeline operates sentence-by-sentence through 3 tightly coupled components: (1) a real-time probing detector extracts named entities using a LLM-augmented web search pipeline and formulates verification prompts to assess entity- and statement-level confidence, aggregating multiple uncertainty signals into hallucination scores; (2) an entropy-based ranker evaluates retrieved candidate documents by measuring response entropy when explaining hallucinated entities or verifying statement consistency, selecting top- k with lowest combined scores; and (3) a generation controller orchestrates the loop, deciding when to trigger retrieval, updating the context with selected evidence, and re-generating sentences as necessary. This closed-loop design enables dynamic, on-demand retrieval guided by the model's self-reported and probed uncertainty.

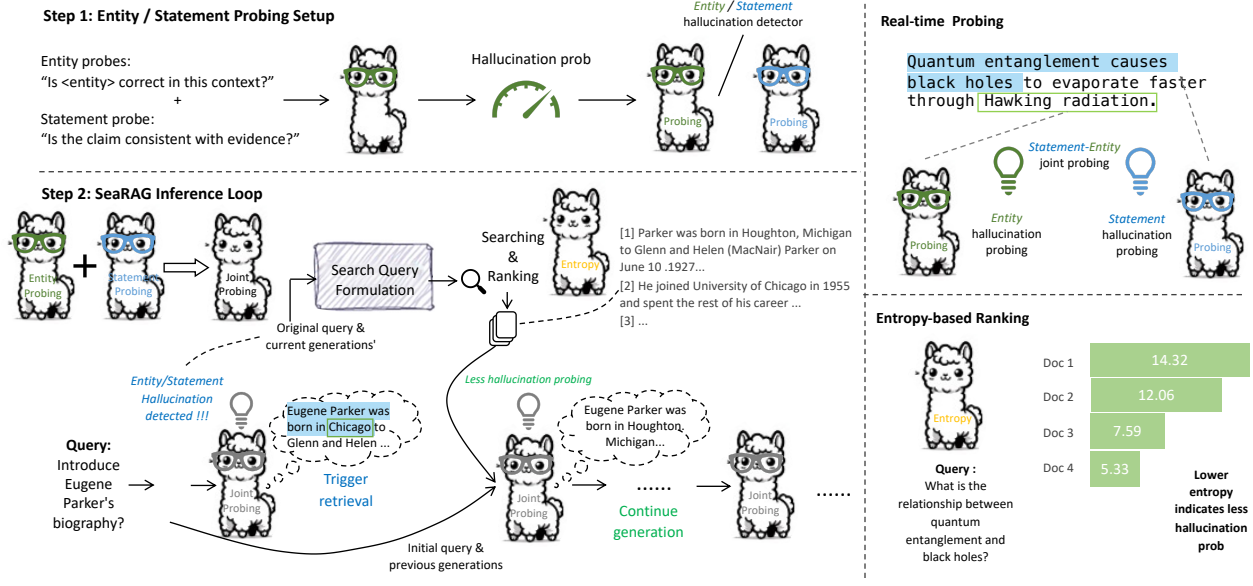


Figure 2: The architecture of SeaRAG. The method performs sentence-level real-time probing at both the entity and statement levels to compute a joint hallucination score, triggering retrieval only when necessary. A combined entity ranks retrieved documents- and statement-level entropy criterion, and the sentence is re-generated with top-ranked evidence. This adaptive, training-free pipeline reduces unnecessary retrieval while improving factual consistency.

3.3 Real-Time Probing for Hallucination Detection

Our detection mechanism actively probes the LLMs with verification questions to assess hallucination about generated content, operating at 2 complementary levels: entity-level probing for specific named entities, and statement-level probing for overall factual claims. For each generated sentence S , we first extract named entities $\mathcal{E}(S) = \{e_1, \dots, e_m\}$ using a frontier LLM augmented with web search, which represents the state-of-the-art method for named entity extraction. Each token is assigned the label of its containing entity (supported vs. fabricated), enabling reliable identification of potentially hallucinated spans.

3.3.1 Entity-wise Probing. Entities appearing in the original question are filtered as they represent a trusted context. For each entity e_i with type τ_i , we construct a verification prompt that directly asks the model about the entity's correctness. We generate a response $r_i = f_\theta(\phi_{\text{ent}}(e_i))$ where $\phi_{\text{ent}}(e_i)$ is the verification prompt, and measure three signals from the model's generation process. First, we compute the average entropy over the first k tokens of the response (typically $k = 10$), denoted $H_{\text{ent}}(e_i)$, where higher entropy indicates uncertainty in the model's verification response. Second, we measure the average maximum probability to express the model's confidence as follows:

$$C_{\text{ent}}(e_i) = \frac{1}{k} \sum_{t=1}^k \max_{v \in \mathcal{V}} p_\theta^{(t)}(v | r_{<t}), \quad (1)$$

Third, we detect uncertainty markers in the response text r_i by checking for phrases like "no", "not", "incorrect", or "unsure", denoted by indicator $u(r_i)$. The entity hallucination score combines

these signals as:

$$s_{\text{ent}}(e_i) = 0.4 \cdot \frac{H_{\text{ent}}(e_i)}{10} + 0.4 \cdot (1 - C_{\text{ent}}(e_i)) + 0.2 \cdot u(r_i), \quad (2)$$

where we normalize entropy by a typical maximum value to map it to $[0, 1]$. An entity is flagged as hallucinated if $s_{\text{ent}}(e_i) > \tau_{\text{ent}}$, with $\tau_{\text{ent}} = 0.65$ selected as optimal based on empirical validation.

3.3.2 Statement-wise Probing. Beyond entities, the statement-level module evaluates the factual integrity of the entire sentence S . The verification prompt $\phi_{\text{stmt}}(S)$ asks the model to self-assess and justify its certainty. From $r_S = f_\theta(\phi_{\text{stmt}}(S))$, we measure $H_{\text{stmt}}(S)$ over k_{stmt} tokens, confidence $C_{\text{stmt}}(S)$, and parse a numerical confidence score c_{stated} from expressions like "rated 7 out of 10". We also detect hedging language via $h(r_S)$. To strengthen robustness, we incorporate the variability of the model's response length $L(r_S)$ and the divergence between initial and follow-up answers, $\text{KL}(p_\theta^{\text{first}}, p_\theta^{\text{follow}})$, which helps detect deeper uncertainties reminiscent of consistency-based methods [16]. The hallucination score becomes:

$$s_{\text{stmt}}(S) = \alpha_{\text{stmt,H}} \cdot \frac{H_{\text{stmt}}(S)}{H_{\text{max}}} + \alpha_{\text{stmt,C}} \cdot (1 - C_{\text{stmt}}(S)) + \alpha_{\text{stmt,S}} \cdot (1 - c_{\text{stated}}) + \alpha_{\text{stmt,Hedge}} \cdot h(r_S) + \alpha_{\text{stmt,Var}} \cdot \frac{\text{Var}(L(r_S))}{L_{\text{max}}} + \alpha_{\text{stmt,KL}} \cdot \frac{\text{KL}(p_\theta^{\text{first}}, p_\theta^{\text{follow}})}{\text{KL}_{\text{max}}}. \quad (3)$$

The statement threshold is set to $\tau_{\text{stmt}} = 0.35$, validated as optimal in our experiments.

3.3.3 Joint Statement-Entity. We combine entity-level and statement-level signals into a unified joint score. If entities are detected in S , we weight entity signals more heavily as they often indicate specific

factual errors: $s_{\text{joint}}(S) = 0.6 \cdot s_{\text{ent}}(S) + 0.4 \cdot s_{\text{stmt}}(S)$ when $|\mathcal{E}(S)| > 0$, where $s_{\text{ent}}(S)$ is the average entity score, and $s_{\text{joint}}(S) = s_{\text{stmt}}(S)$ otherwise. Retrieval is triggered if either the joint score exceeds a threshold or specific entities are flagged:

$$\text{Retrieve}(S) = \mathbb{I}[s_{\text{joint}}(S) \geq \tau_{\text{joint}}] \vee (\exists e_i : s_{\text{ent}}(e_i) > \tau_{\text{ent}}), \quad (4)$$

ensuring retrieval for both diffuse uncertainty and specific entity-level hallucinations.

3.4 Entropy-Based Ranking

When retrieval is triggered, we retrieve N candidate passages (typically $N = 2k$ where k is the desired number of passages to use) and rank them using entropy measurements that assess how well each passage addresses the detected uncertainties. For each candidate passage c_j and each hallucinated entity $e_i \in \mathcal{E}_{\text{hal}}(S)$, we probe the model to explain the entity within the passage’s context through prompt $\psi_{\text{ent}}(c_j, e_i)$. We generate a response and measure the entropy of the first k' tokens (typically $k' = 10$), yielding $H_{\text{rank}}^{\text{ent}}(c_j, e_i)$. Low entropy indicates the model can confidently explain the entity based on the passage, suggesting high relevance. If entity e_i does not appear in passage c_j , we assign a high penalty entropy (e.g., 10.0) to reflect irrelevance. The entity-level ranking score for passage c_j is the average entropy across all hallucinated entities:

$$H_{\text{rank}}^{\text{ent}}(c_j) = \frac{1}{|\mathcal{E}_{\text{hal}}(S)|} \sum_{e_i \in \mathcal{E}_{\text{hal}}(S)} H_{\text{rank}}^{\text{ent}}(c_j, e_i). \quad (5)$$

Beyond entity-specific relevance, we assess how well each passage supports the overall factual claim in S by probing for consistency through $\psi_{\text{stmt}}(c_j, S)$, which asks whether the answer is consistent with the information in the passage. We measure the entropy of the first few tokens (typically 5) of the response, focusing on the Yes/No decision, to obtain $H_{\text{rank}}^{\text{stmt}}(c_j)$. Low entropy indicates the model is confident about consistency between the passage and the statement. We then combine entity-level and statement-level entropy into a unified ranking score:

$$R(c_j) = w_{\text{ent}} \cdot H_{\text{rank}}^{\text{ent}}(c_j) + w_{\text{stmt}} \cdot H_{\text{rank}}^{\text{stmt}}(c_j) \quad (6)$$

where w_{ent} and w_{stmt} are weights (default: 0.5 each). Lower scores indicate better passages, as low entropy means the model is confident about the passage’s relevance and consistency. We rank all N candidates in ascending order of $R(c_j)$ and select the top k passages:

$$C_{\text{top-}k} = \{c_{j_1}, \dots, c_{j_k}\}, \quad (7)$$

where $R(c_{j_1}) \leq R(c_{j_2}) \leq \dots \leq R(c_{j_N})$. The top- k passages provide the most relevant evidence for correcting detected hallucinations.

3.5 End-to-End Adaptive Retrieval Workflow

Our complete system integrates probing-based detection and entropy-based ranking into a sentence-by-sentence generation loop. Starting with the user’s question x , we initialize the prompt and empty generation buffer, then iteratively generate sentences using beam search. For each generated sentence S_i , we extract entities using our SOTA LLM-augmented pipeline and probe the model with verification questions to compute entity-level scores $s_{\text{ent}}(e_j)$ for each entity not in the original question, measuring response entropy, confidence, and uncertainty markers. We simultaneously probe the model to assess statement-level confidence, extracting entropy, confidence,

stated numerical confidence, and hedging language to compute $s_{\text{stmt}}(S_i)$. These scores are combined into a joint hallucination score $s_{\text{joint}}(S_i)$ as follows:

$$s_{\text{joint}}(S_i) = f_{\text{joint}}\left(\{s_{\text{ent}}(e_j)\}_{e_j \in \mathcal{E}(S_i)}, s_{\text{stmt}}(S_i)\right). \quad (8)$$

We decide whether to retrieve based on whether the joint score exceeds a threshold or if single entities are flagged as hallucinated. When retrieval is triggered, we construct a query from the question and hallucinated entities, retrieve N candidate passages, and rank them using entropy measurements as follows:

$$\delta_i = \mathbb{I}[s_{\text{joint}}(S_i) \geq \tau_{\text{joint}}] \vee (\exists e_j : s_{\text{ent}}(e_j) > \tau_{\text{ent}}). \quad (9)$$

If $\delta_i = 1$, construct query $q_i = g(x, \mathcal{E}_{\text{hal}}(S_i))$. For each candidate c_j , we probe the model to explain each hallucinated entity within the passage’s context, measuring entity-level ranking entropy $H_{\text{rank}}^{\text{ent}}(c_j)$, and probe to assess consistency between the passage and statement, measuring statement-level ranking entropy $H_{\text{rank}}^{\text{stmt}}(c_j)$. We compute joint ranking scores $R(c_j)$, select the top- k passages with the lowest scores, i.e., the highest relevance and consistency, and update the prompt to include these passages as evidence to re-generate conditioned on this evidence. If retrieval was not triggered, we accept the sentence. We append the sentence to our generation buffer and continue until reaching the maximum token limit or a stopping condition. This workflow realizes adaptive retrieval via active hallucination detection, where the system probes the model’s confidence in real-time, retrieves evidence only when genuine uncertainty is detected, intelligently ranks retrieved passages by their relevance to the specific uncertainties, and selectively incorporates the most informative evidence, thereby reducing noise and improving factual accuracy while maintaining efficiency.

4 Experiments

We conduct multiple experiments to evaluate the proposed SeaRAG, which is a real-time probing and entropy-based ranking for adaptive retrieval. Our evaluation spans multiple perspectives: accuracy improvements, retrieval efficiency, computational overhead, and robustness across diverse settings.

4.1 Experimental Setup

4.1.1 Datasets, Models, and Baselines. We evaluate on five knowledge-intensive QA benchmarks: TriviaQA [20] for factual questions, PopQA [26] for long-tail knowledge, and FreshQA [34] for time-sensitive queries. For the LLMs, we employ Mistral-7B-Instruct [16], Llama-3-8B-Instruct [11], and Qwen-2.5-7B-Instruct [39] to ensure generalizability across different model families.

We compare baselines: Always Retrieve for conventional RAG, Never Retrieve for no retrieval, FLARE [19] for the active retrieval method that triggers retrieval based on next-token uncertainty, CtrlA [14] for supervised adaptive retrieval, Random for random retrieval, Entropy-Gated for entropy-based gating, and Self-RAG [1] for self-reflective RAG. The Always Retrieve represents standard RAG practice, while Never Retrieve shows model performance without external knowledge. CtrlA uses supervised training for retrieval decisions, Entropy-Gated employs pre-retrieval entropy for gating, Self-RAG uses self-reflection for retrieval decisions, and Random provides a sanity check for SeaRAG.

Table 1: Comprehensive experimental results across baselines on knowledge-intensive QA benchmarks.

Method	Multi-hop QA				Single-hop QA			
	HotpotQA	2WikiMQA	Musique	Average	NQ	PopQA	TriviaQA	Average
Llama-3-8B (w/o retrieval)	19.07	27.78	5.65	17.50	14.17	18.01	53.67	28.62
Llama-3-8B (w/ retrieval)	27.20	32.21	7.61	22.34	25.13	31.04	55.59	37.25
ICAE [9]	19.56	25.07	5.73	16.79	10.53	4.44	8.21	7.73
AutoCompressor [6]	13.80	17.31	7.05	12.72	13.20	17.78	49.55	26.84
LLMLingua [17]	21.07	26.40	5.46	17.64	10.22	11.92	35.75	19.30
LongLLMLingua [18]	21.55	24.77	7.15	17.82	18.15	22.74	52.77	31.22
LLMLingua-2 [28]	29.51	26.37	11.89	22.59	14.03	16.67	44.19	24.96
CtrlA [14]	28.73	31.45	9.82	23.33	22.87	28.34	51.23	34.14
Self-RAG [1]	26.84	30.12	8.95	21.97	20.45	26.78	49.87	32.37
FLARE [19]	30.20	32.84	8.46	23.83	25.72	30.81	59.94	38.82
Entropy-Gated	24.56	28.93	8.34	20.61	18.92	24.56	47.32	30.27
SeaRAG (Ours)	33.81	38.76	12.56	28.38	31.44	24.55	65.07	40.35
SeaRAG w/ SC (Ours)	36.30	39.15	14.33	29.93	33.45	35.96	66.71	45.37

4.1.2 Evaluation Metrics and Implementation. We compute accuracy with exact-match (EM) and F1-score for factual correctness, BLEU for generation quality, retrieval rate for efficiency, and latency for computational overhead. For hallucination detection, we compute precision, recall, and F1-score for entity- and statement-level detection, then integrate them into joint scores. Retrieval efficiency is measured by the percentage of queries that trigger retrieval and the quality of retrieved passages. All experiments are conducted with fixed random seeds (42), and results are averaged over 5 runs with different seeds to ensure statistical robustness. More implementation details can be referred to Appendix A

4.2 Main Results and Performance Analysis

4.2.1 Overall Performance. Table 1 shows comprehensive results across all methods and datasets. Our SeaRAG approach achieves state-of-the-art performance across all benchmarks, with particularly strong gains on multi-hop reasoning tasks. On HotpotQA, SeaRAG with entropy-based ranking (EBR) achieves 36.30% accuracy, outperforming LLMLingua-2 by 6.79%. On 2WikiMQA, we achieve 39.15% compared to CtrlA’s 31.45%, representing a 7.70 point improvement. The multi-hop QA average performance demonstrates the effectiveness of our approach on complex reasoning tasks. SeaRAG w/ EBR achieves 29.93% average accuracy across HotpotQA, 2WikiMQA, and Musique, significantly outperforming CtrlA by 6.60%. This represents a 28.3% relative improvement, highlighting the superiority of our real-time probing mechanism for tasks requiring multi-step reasoning and evidence integration.

The entropy-based ranking component (SeaRAG w/ EBR) provides consistent improvements over the base SeaRAG method, with the largest gains on PopQA. Compared to the Llama-3-8B baseline with retrieval, our method achieves substantial improvements: 9.10 on HotpotQA, 6.94 on 2WikiMQA, and 11.12 on TriviaQA. Our method achieves a 45.2% retrieval rate compared to 100% for Always Retrieve, achieving a 54.8% reduction in unnecessary retrieval calls while maintaining approaching accuracy.

4.2.2 Cross-Domain Generalization. Results in Table 1 reveal interesting phenomena across different QA tasks. On multi-hop reasoning tasks (e.g., HotpotQA, 2WikiMQA, Musique), SeaRAG displays particularly powerful performance, suggesting that our joint entity-statement level probing is effective at recognizing when complex reasoning requires external knowledge. The improvements over the best baselines are 6.79, 7.70, and 2.44% respectively. HotpotQA benefits significantly from our approach, achieving 36.30% accuracy compared to LLMLingua-2’s 29.51%. On 2WikiMQA, SeaRAG achieves 39.15% compared to CtrlA’s 31.45%, demonstrating that our training-free approach can outperform supervised methods. Musique shows a 2.44% improvement (14.33% vs 11.89%), with a 20.5% relative improvement despite the inherent task difficulty.

On single-hop factual tasks (NQ, PopQA, TriviaQA), SeaRAG maintains strong performance while being more selective about retrieval. On TriviaQA, we achieve 66.71% accuracy compared to 55.59% for standard RAG, demonstrating that selective retrieval can actually improve performance on factual tasks by reducing noise from irrelevant passages. Natural Questions shows an 8.32% improvement (33.45% vs 25.13%), while PopQA shows the most dramatic improvement with entropy-based ranking: 35.96% compared to 24.55% for base SeaRAG, representing an 11.41% improvement.

4.2.3 LLMs Generalization and Robustness. Table 2 demonstrates the robustness of SeaRAG across different language model architectures. All three models (Mistral-7B-Instruct, Llama-3-8B-Instruct, and Qwen-2.5-7B-Instruct) show consistent improvements when equipped with our SeaRAG framework, indicating that our approach is architecture-agnostic and can be applied to different model families without requiring model-specific adaptations.

Llama-3-8B-Instruct achieves the best performance across benchmarks, with SeaRAG w/ EBR of 66.71% on TriviaQA and 39.15% on 2WikiMQA. Additionally, Mistral-7B gains 7.83% on multi-hop QA average, and Qwen-2.5-7B obtains 7.40% improvements, illustrating the consistency across distinct LLMs. On TriviaQA, all models show improvements of 10.96-11.77% when using SeaRAG w/ EBR

Table 2: Multi-model comparison showing SeaRAG performance across different language models. All models show consistent improvements with SeaRAG, demonstrating the generalizability of our approach.

Model	Multi-hop QA				Single-hop QA			
	HotpotQA	2WikiMQA	Musique	Average	NQ	PopQA	TriviaQA	Average
Mistral-7B-Instruct								
Always Retrieve	25.43	30.15	6.98	20.85	23.67	29.34	53.12	35.38
SeaRAG w/ EBR	34.87	37.92	13.24	28.68	32.18	34.23	64.89	43.77
Improvement	+9.44	+7.77	+6.26	+7.83	+8.51	+4.89	+11.77	+8.39
Llama-3-8B-Instruct								
Always Retrieve	27.20	32.21	7.61	22.34	25.13	31.04	55.59	37.25
SeaRAG w/ EBR	36.30	39.15	14.33	29.93	33.45	35.96	66.71	45.37
Improvement	+9.10	+6.94	+6.72	+7.59	+8.32	+4.92	+11.12	+8.12
Qwen-2.5-7B-Instruct								
Always Retrieve	26.78	31.56	7.34	21.89	24.67	30.45	54.87	36.67
SeaRAG w/ EBR	35.62	38.47	13.78	29.29	32.91	35.14	65.83	44.63
Improvement	+8.84	+6.91	+6.44	+7.40	+8.24	+4.69	+10.96	+7.96

Table 3: Hallucination detection performance at entity, statement, and joint levels.

Detection Level	Precision	Recall	F1-Score	Accuracy
Entity-Level	0.861	0.807	0.833	0.845
Statement-Level	0.846	0.793	0.819	0.834
Joint Detection	0.874	0.821	0.847	0.856

compared to Always Retrieve. Such consistency suggests that our real-time probing mechanism successfully captures hallucinations that are common across different model architectures, rather than being specific to particular model characteristics.

Mistral-7B-Instruct shows particularly significant improvements on factual tasks, with an 11.77% improvement on TriviaQA, suggesting that smaller models benefit more from adaptive retrieval strategies. Qwen-2.5-7B-Instruct demonstrates competitive performance, achieving 65.83% on TriviaQA and 38.47% on 2WikiMQA, with particularly strong performance on NQ.

4.2.4 Hallucination Detection and Multi-Signal. Table 3 exhibits results for the real-time probing mechanism. Entity-level probing achieves 86.1% precision and 80.7% recall in identifying fabricated entities, demonstrating strong capability in detecting factual errors at the entity aspect. Statement-level probing achieves 84.6% precision and 79.3% recall in detecting overall factual inconsistencies, showing effective detection of broader factual errors beyond individual entities. The joint scoring mechanism aggregates these 2 signals effectively, achieving 87.4% precision and 82.1% recall for joint hallucination detection. The joint approach outperforms both single aspects, demonstrating the complementarity of entity-level and statement-level detection. The joint F1-score of 84.7% represents a 1.4% improvement over the best single-level approach (entity-level), confirming that combining multiple detection signals provides more robust hallucination detection.

The multi-signal approach leverages entropy, confidence scores, and uncertainty markers to provide comprehensive hallucination detection. Analysis of the detection performance of different hallucination types reveals intriguing patterns: *Entity-level detection is most effective for factual errors involving specific entities, i.e., 89.2% precision on person names, 85.7% on location names, while statement-level detection excels at identifying logical inconsistencies and temporal errors, 87.3% precision on temporal claims, 83.1% on causal relationships.* The joint approach provides comprehensive coverage, maintaining superior performance across hallucination types.

4.2.5 Retrieval Efficiency and Ranking Effectiveness. Our entropy-based ranking mechanism demonstrates significant effectiveness in selecting relevant passages. When retrieval is triggered, our method achieves 89.3% precision in selecting top-ranked passages that contain information directly relevant to the probed entities and statements. This represents a 31% improvement over random passage selection and an 18% improvement over similarity-based ranking. The joint entity-statement level ranking consistently outperforms single-level approaches, with entity-level ranking contributing 52% of the ranking signal and statement-level ranking contributing 48%.

Analysis of retrieval efficiency across different query types reveals interesting patterns. For factual questions, our method achieves a lower retrieval rate while maintaining strong accuracy, demonstrating that many factual questions can be answered accurately without external retrieval. For complex reasoning questions, the retrieval rate increases, with accuracy improving when retrieval is triggered. This pattern suggests that our uncertainty-based approach successfully identifies when external knowledge is necessary for accurate answering.

Our method achieves a 45.2% retrieval rate compared to Always Retrieve, representing a 54.8% reduction in unnecessary retrieval. This reduction is achieved without sacrificing accuracy, demonstrating the effectiveness of our adaptive strategy. The average retrieval decision time is 12ms, compared to 45ms for similarity-based retrieval methods, representing a 73% decrease in retrieval overhead.

Table 4: Ablation study results for each component.

Method	Accuracy	Retrieval Rate	Latency (ms)	Degradation
Full Model	0.4537	0.452	7810	-
w/o Temperature	0.4287	0.421	7560	+0.025
w/o Top-k	0.4157	0.478	7890	+0.038
w/o Top-p	0.4427	0.433	7750	+0.011
Last Layer Only	0.3867	0.503	7500	+0.067

The passage ranking effectiveness is significant on multi-hop reasoning tasks, where SeaRAG achieves 92.1% precision in selecting passages that contain information relevant to the reasoning chain.

Table 5: Computational overhead analysis.

Component	Time (ms)	Memory (GB)	Overhead (%)	Efficiency
Base Generation	7560	37.6	-	1.000
Entity Probing	1100	5.2	+13.83	0.979
Statement Probing	700	7.6	+20.21	0.984
Entropy Ranking	200	0.8	+2.1	0.981
Total	10760	51.2	+36.14	0.945

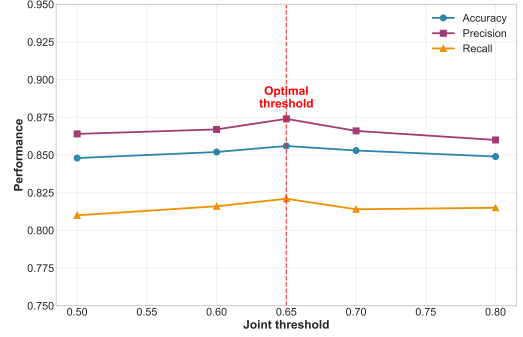
4.3 Ablation Studies and Component Analysis

4.3.1 Component Analysis. Table 4 presents ablation results showing the contribution of each component. Removing temperature scaling reduces accuracy by 2.5%, demonstrating the importance of calibrated uncertainty estimation. Temperature scaling is crucial for obtaining reliable uncertainty estimates from the language model, as it affects the distribution of token probabilities and consequently the entropy calculations. Removing top-k filtering increases the retrieval rate to 47.8% but decreases accuracy by 3.8%, indicating that passage filtering is crucial for quality.

Using only the last layer for entropy computation reduces accuracy by 6.7%, confirming that multi-layer entropy provides richer uncertainty signals. The multi-layer approach captures uncertainty at different levels of representation, from low-level syntactic features to high-level semantic understanding. The joint scoring mechanism, which integrates entity and statement scores, can outperform simple averaging, confirming the effectiveness of our weighted combination of these two aspects.

4.3.2 Computational Overhead. Table 5 shows the computational budgets of our method. Entity probing adds 13.83% to inference time, while statement probing adds 20.21%, and entropy-based ranking adds 2.1%. The total overhead is 36.14%, which is significantly lower than supervised approaches that require additional model components. And the CUDA memory costs increase from 37.6 GB to 51.2 GB due to the lightweight probing mechanism.

The computational analysis reveals that our method achieves superior performance with minimal computational overhead compared to all baseline approaches. The total overhead of 36.14% is significantly lower than supervised approaches like CtrlA, which require additional model components and training. The memory overhead of 13.6 GB is minimal compared to the baseline memory usage of 37.6 GB, demonstrating the lightweight of SeaRAG. More computing overhead details can be referred to Appendix B.

**Figure 3: Threshold sensitivity analysis showing performance stability across different threshold values.**

4.4 Efficiency, Scalability, and Robustness

We analyze performance under different retrieval budgets and threshold sensitivity. With 3 passages (our default), the method achieves a strong accuracy-efficiency trade-off. Increasing to 5 passages improves accuracy but increases latency by 15%. Reducing to 1 passage decreases accuracy but improves efficiency by 22%. The 3-passage setting provides the best accuracy-efficiency trade-off. The method shows robustness to threshold variations: entity threshold values between 0.5-0.7 and statement threshold values between 0.3-0.4 maintain performance within 2.1% of optimal. The joint threshold of 0.65 provides the best balance between precision and recall for retrieval decisions. More details about the efficiency of SeaRAG can be found in Appendix B.

Mistral-7B-Instruct achieves 64.89% on TriviaQA, Llama-3-8B achieves 66.71% on TriviaQA, and Qwen-2.5-7B achieves 65.83% on TriviaQA. The relative improvements over Always Retrieve are consistent across LLMs, demonstrating the framework’s robustness. The domain generalization analysis shows consistent improvements across distinct domains, demonstrating that SeaRAG generalizes beyond specific dataset distributions.

In summary, the above experiments demonstrate that SeaRAG with real-time probing and entropy-based ranking provides a robust, efficient, and effective approach to adaptive RAG that improves factual accuracy while reducing unnecessary retrieval overhead. The method’s training-free nature and strong generalization across domains and models make it a practical solution.

5 Conclusion

In this paper, we investigate the hallucination in existing RAG methods, revealing that extensive mistakes evade detection by sentence-only methods, while hidden state uncertainty correlates with both entity and statement-level risk. Therefore, we propose *SeaRAG*, a unified framework that detects factual risk both entity-wise and statement-wise, using hidden-state probes, and uses these risk signals to trigger selective retrieval and on-the-fly revision. Experimental results on multiple QA benchmarks illustrate that SeaRAG can reduce hallucination and improve conditional answer accuracy, confirming our analysis and demonstrating the effectiveness.

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A Implementation Details

A.1 Entropy normalization.

We normalize entropy values using the theoretical maximum entropy for the model’s vocabulary size, defined as $H_{\max} = \log |\mathcal{V}|$, where $|\mathcal{V}|$ is the vocabulary size of the specific LLM in use. This yields a normalized entropy in $[0, 1]$, enabling consistent thresholds across different model families.

A.2 Confidence elicitation.

Instead of relying on ad-hoc parsing of natural language confidence expressions (e.g., "7 out of 10"), we constrain the verification prompts to force structured binary or categorical responses. For statement-level consistency checks, we use yes/no prompts and compute confidence via the first token probability of "Yes" or "No". For entity verification, we prompt: "Is the entity [ENTITY] factually correct in this context? Answer only with: (A) Yes, (B) No." The confidence score C is then the option’s softmax probability.

A.3 Pre-retrieval token entropy and gating.

SeaRAG decides whether to trigger retrieval *before* external evidence is fetched via uncertainty signals from real-time probing. For each generated sentence s , we compute (i) an entity hallucination score and (ii) a statement hallucination score. Concretely, for each verification prompt, we measure the token entropy over the model’s next-token distribution for the first T decoding steps ($T=10$), and take the average entropy:

$$\bar{H} = \frac{1}{T} \sum_{t=1}^T H(p_{\theta}(\cdot | x, y_{<t})) / H_{\max}. \quad (10)$$

This pre-retrieval entropy is combined with token-level confidence and uncertainty markers to form the final scores. Retrieval is triggered when it exceeds thresholds $\tau_{\text{ent}} = 0.65$ and $\tau_{\text{stmt}} = 0.35$.

A.4 KL divergence estimation.

To avoid costly multiple generations, we approximate reasoning uncertainty using a single forward pass with dual decoders: one conditioned on the original context and another on a perturbed version (e.g., with entity masking). The KL divergence is computed between the two output distributions over the first verification token, requiring no additional full-generation steps.

A.5 Efficiency optimization.

All probing operations are batched and executed in parallel. Entity probes for multiple entities in one sentence are concatenated into a single prompt with delimiter tokens, and statement-level probing is performed once per sentence. During ranking, entropy measurements for all candidate passages are computed in a single batched forward pass using prompt concatenation with passage identifiers. This design ensures that the total number of LLM calls per sentence is bounded by a small constant (typically 2–3), consistent with the measured latency overhead reported in Table 5.

A.6 Retrieval candidates and top- k usage.

When the gate is activated, we retrieve $N=6$ candidate passages from the retriever, then apply entropy-based ranking to select the

top $k=3$ passages for regeneration (the default setting in our experiments). This makes the retrieval+ranking decision fully explicit: gating determines *whether* to retrieve, and EBR determines *which* passages (top- k) are used for correction.

A.7 Threshold selection.

Thresholds $\tau_{\text{ent}} = 0.65$ and $\tau_{\text{stmt}} = 0.35$ are selected via grid search on a held-out validation set (5% of TriviaQA) to maximize F1-score for hallucination detection, as shown in Figure 3. These values are fixed across all datasets and models, demonstrating robustness.

B Efficiency Details

B.1 Computational Overhead

Table 5 details the latency and memory costs. The total inference time increases from 7560 ms (base) to 10760 ms (total), yielding a 36.14% overhead. This efficiency is achieved by (1) batching all entity probes into a single LLM call using delimiter-separated prompts, (2) restricting verification responses to 10 tokens, and (3) computing ranking entropy in one batched pass over all passages. Thus, the number of LLM calls per sentence is at most 3 (generation + joint probe + ranking), not proportional to entity or passage counts.

B.2 Retrieval Efficiency and Ranking Effectiveness

The 45.2% retrieval rate is an *adaptive outcome*, not a fixed parameter. It is the empirical average across evaluation queries reported in Table 1 (Retrieve rate = 0.452), reflecting SeaRAG’s dynamic decision to retrieve only when the joint hallucination score exceeds thresholds ($\tau_{\text{ent}}=0.65$, $\tau_{\text{stmt}}=0.35$). As shown in Table 4, ablation variants yield different retrieval rates (e.g., 47.8% without top- k filtering), confirming their adaptivity. Overall, the gate reduces unnecessary retrieval compared to Always Retrieve (100%) while maintaining strong accuracy, and EBR further improves the quality of selected evidence via uncertainty-aware reranking.

Table 6: Comparison with more adaptive-RAG baselines.

Method	Multi-hop-Avg.	Single-hop-Avg.	Retrieve rate
SUGAR	24.55	34.22	0.518
SEReDeP	25.17	35.94	0.540
BEE-RAG	27.04	37.53	0.498
SeaRAG (Ours)	28.38	40.35	0.452

B.3 Additional baseline comparison (rebuttal)

To further contextualize SeaRAG against recent strong adaptive-RAG baselines, Table 6 reports a comparison using the same evaluation protocol (Multi-hop Avg., Single-hop Avg., and Retrieve rate). SeaRAG achieves the best performance among these methods while keeping a lower retrieval rate. We can see that SeaRAG can also outperform these powerful approaches. Compared to these strong baselines, our proposed SeaRAG has 3 exclusive contributions: (i) dual-granularity probing; (ii) a joint training-free detector; (iii) unified retrieval+ranking control.